

Horizon-Scaled Saturation and Rare-Event Formation of Early Supermassive Black Holes

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Abstract

This work develops a phenomenological cosmological prediction arising from a statistical reformulation of spacetime in which gravity corresponds to a local deficit in accessible null solid angle. In this framework, black hole formation is interpreted as a kinematic saturation event occurring when the coarse-grained null-directional strain exceeds a critical threshold. This naturally yields a rare-event formation channel for supermassive black holes (SMBHs) in the early universe, governed by the extreme tail of primordial fluctuation statistics rather than by prolonged Eddington-limited accretion.

The model predicts a redshift-dependent kinematic support boundary set by the Hubble-horizon mass scale,

$$M_H(z) = \frac{c^3}{GH(z)},$$

together with a threshold-crossing probability that admits a minimal Gaussian-tail realization. Reported high-redshift black-hole candidates, including objects near $z \sim 10$, can be interpreted as extreme-tail formation events occurring well within this absolute kinematic boundary. While $M_H(z)$ provides a theoretical ceiling, the operative observational boundary is a much lower rare-event capacity envelope, $M_{\text{env}}(z)$, controlled by the fluctuation variance. Because the calibrated envelope is itself a conservative upper-capacity construction, isolated mild exceedances would place the model under tension. By contrast, a statistically robust population of high-redshift black hole mass estimates lying above the predicted envelope after propagation of the quoted calibration band and observational mass-estimate uncertainties would falsify the Gaussian-tail saturation-threshold realization under the adopted single-anchor calibration.

This provides a concrete, statistically grounded observational discriminator between null-kinematic spacetime and standard accretion-driven structure formation without modifying tested gravitational dynamics.

1 Introduction

The geometric description of gravity provided by General Relativity (GR) has withstood all experimental tests in weak- and strong-field regimes. Nevertheless, the interpretation of spacetime as a fundamental continuum leads to conceptual tensions at singularities and horizons, where geometric invariants diverge while physical evolution becomes ill-defined. This motivates the exploration of formulations in which geometry is emergent rather than fundamental.

In Refs. [1, 2], spacetime is reformulated as a statistical state of locally constrained null propagation, with the latter providing a broader constitutive development of the strain sector. The primitive object is not a metric tensor but a local distribution of null directions,

$$P(\hat{\mathbf{k}}; x),$$

defined on the unit sphere with $|\mathbf{k}| = c$. In the absence of structure, this distribution is isotropic, corresponding to maximal directional entropy and no preferred transport.

Gravity, in this framework, corresponds to a local *deficit in accessible null solid angle*. Curvature is reinterpreted statistically as a deformation of the directional ensemble away from isotropy. The effective macroscopic transport velocity $\mathbf{v}(\mathbf{x}, t)$ arises as the first angular moment of this distribution,

$$\mathbf{v}(x) = c \int \hat{\mathbf{k}} P(\hat{\mathbf{k}}; x) d\Omega,$$

and is therefore not fundamental but derived from the underlying directional statistics.

In the broader constitutive development of the NDK program, this same null-directional distribution also defines higher angular moments and an associated scalar strain functional, so that the geometric solid-angle deficit used here should be understood as the minimal phenomenological expression of a more general informational strain in the local directional ensemble, and the emergent transport velocity should be understood as one element of a larger hierarchy of coarse-grained observables rather than as the sole dynamical descriptor.

The degree of residual directional freedom is quantified by an indeterminacy parameter χ , which measures the remaining variance of the null ensemble. As the local distribution becomes increasingly polarized (i.e., concentrated into a narrow angular sector), the accessible solid angle shrinks and χ decreases. The limit in which the distribution becomes fully polarized corresponds to vanishing directional variance and defines a *kinematic saturation boundary*.

In this statistical picture, the event horizon is not merely a geometric null hypersurface but the physical limit at which the local null ensemble exhausts its directional degrees of freedom. The mathematical successes of GR are preserved in regimes where the directional distribution varies smoothly and remains far from saturation, but the ontological interpretation differs: curvature reflects constrained null-direction statistics rather than deformation of a fundamental manifold.

The present paper focuses on phenomenological consequences of this statistical reinterpretation, deriving cosmological observational signatures that distinguish null-directional saturation from standard geometric collapse mechanisms—with particular emphasis on the implications for early-universe black hole formation.

2 Theoretical Primitives

This section summarizes the phenomenological variables induced by the null-directional distribution $P(\hat{\mathbf{k}}; x)$ introduced above; detailed construction is given in Refs. [1, 2].

At the broader constitutive level, one may regard the local null ensemble as defining a hierarchy of coarse-grained descriptors,

$$P(\hat{\mathbf{k}}; x) \longrightarrow \{\mathbf{v}(x), \chi(x), I[P](x), \dots\},$$

where $\mathbf{v}$ encodes the first angular moment, χ the residual directional freedom, and $I[P]$ a scalar measure of directional ordering relative to isotropy. The present paper uses only the minimal subset of this hierarchy needed to formulate a saturation threshold and its cosmological consequences.

2.1 Directional Moment and Effective Transport

The emergent spacetime transport velocity $\mathbf{v}(x)$ is not fundamental but corresponds to the first angular moment of the local null distribution,

$$\mathbf{v}(x) = c \int \hat{\mathbf{k}} P(\hat{\mathbf{k}}; x) d\Omega. \tag{1}$$

An isotropic distribution yields $\mathbf{v} = 0$, while increasing polarization of $P(\hat{\mathbf{k}}; x)$ produces a nonzero mean transport direction.

Thus, the emergent transport vector encodes directional polarization of the null ensemble rather than motion of an underlying geometric medium.

2.2 Directional Variance and Indeterminacy

The capacity for temporal evolution is quantified by the residual angular variance of the null ensemble. This is summarized phenomenologically by the indeterminacy parameter

$$\chi = \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2}}, \quad (2)$$

which measures the fraction of directional degrees of freedom remaining after polarization.

In statistical terms, χ vanishes when the null distribution becomes fully polarized into a delta-function on the sphere, corresponding to zero accessible solid angle. Standard relativistic time dilation corresponds to the reduction of χ as the directional ensemble becomes increasingly anisotropic.

2.3 Kinematic Saturation

Because $P(\hat{\mathbf{k}}; x)$ is defined on the unit sphere, the magnitude of its first moment is bounded by c . The limit

$$\lim_{|\mathbf{v}| \rightarrow c} \chi = 0 \quad (3)$$

defines a *kinematic saturation point*, corresponding to complete polarization of the local null ensemble.

In this limit, accessible directional variance is exhausted; the horizon corresponds to the saturation boundary where the local null solid angle collapses, i.e. the extremal polarization limit of the ensemble.

3 Direct Kinematic Saturation of Primordial SMBHs

Standard cosmological models face increasing tension in explaining the rapid assembly of massive black holes at very high redshift. Recent observations have reported accreting massive black-hole candidates at $z \gtrsim 8$, including candidates near $z \sim 10$, motivating renewed scrutiny of early black-hole growth channels [3–5]. Growing such masses from stellar seeds via accretion is constrained by the Eddington limit [6].

3.1 Mechanism of Direct Collapse

The null-kinematic framework allows for *Direct Kinematic Saturation*. In the radiation-dominated early universe, large-amplitude density fluctuations induce convergent spacetime flow. In the present phenomenological realization, a horizon-scale saturation event is taken to occur once the local flow convergence threshold is crossed, rather than through prolonged hydrostatic collapse.

The characteristic coherent flow velocity scale v_{flow} at a length scale R is bounded by the cosmological expansion rate $H(z)$ through dimensional consistency,

$$v_{\text{flow}} \sim H(z)R. \quad (4)$$

This scaling reflects the coherence scale of primordial velocity flows sourced by density perturbations, rather than the homogeneous background expansion. Saturation occurs when the flow velocity approaches the invariant speed, $v_{\text{flow}} \rightarrow c$. In particular, $H(z)$ should be understood

here as setting an upper bound on the largest causally coherent flow structures available at a given cosmic epoch, not as a driver of inward motion. The convergent flows responsible for saturation arise from primordial density perturbations within this coherence scale, while the homogeneous expansion itself remains kinematically divergent. The null-kinematic mechanism therefore does not identify cosmic expansion as a collapse agent, but uses $H(z)$ purely to characterize the maximum spatial extent over which coherent inflow can develop. Let the causal coherence length of the macroscopic flow be $R_{\text{coh}} = c/H(z)$. Saturation implies the effective kinematic radius approaches a geometric limit analogous to the standard Schwarzschild radius, $r_s = 2GM/c^2$. Equating the saturation radius to the coherence length up to an order-unity geometric factor β , such that $r_{\text{sat}} = \beta R_{\text{coh}} \approx r_s$, yields a characteristic capacity mass:

$$M_H(z) = \frac{\beta c^3}{2GH(z)}. \quad (5)$$

For phenomenological simplicity, all order-unity geometric factors (including β and the numerical factor of 2 relating r_s to GM/c^2) are absorbed into the definition of the baseline horizon-scale capacity mass,

$$M_H(z) \equiv \frac{c^3}{GH(z)}. \quad (6)$$

Thus $M_H(z)$ should be interpreted as a causal kinematic scaling relation defined up to $\mathcal{O}(1)$ geometry-dependent coefficients. The appearance of Newton’s constant G does not assume a standard geometric collapse mechanism. In the null-kinematic description, G functions as an empirical coupling that calibrates the relationship between mass-energy and the strength of directional deformation. The scale $M_H(z)$ should therefore be interpreted as a causal kinematic support scale (up to order-unity geometry) set by causal coherence, not as a dynamical statement about hydrostatic collapse.

3.2 Prediction

This mechanism predicts a population of “primordial saturation zones” (black holes) formed well before the first stars, with a mass function determined by the power spectrum of flow fluctuations rather than stellar evolution. This naturally accommodates the reported high-redshift accreting black-hole candidates identified in recent JWST-era observations [3–5], which are otherwise difficult to reconcile with standard Eddington-limited growth without invoking very massive seeds or prolonged super-Eddington episodes. This mechanism does not preclude subsequent secular accretion; rather, it provides a direct-formation channel whose initial mass scale is set by kinematic saturation rather than by integrated Eddington-limited growth. The theory predicts that the mass distribution of these objects should trace the primordial flow velocity power spectrum, distinguishing them from population-III stellar remnants.

Mathematically, this rare-tail formation channel shares structural similarities with standard Primordial Black Hole (PBH) formation models, which also rely on the threshold-crossing of early primordial perturbations. However, the physical trigger differs fundamentally. Standard PBH formation requires radiation-pressure-overcoming gravitational collapse, typically requiring sufficiently large small-scale power (in many models achieved via features in the primordial spectrum). In contrast, null-kinematic saturation is a direct geometric threshold (the local exhaustion of directional variance) that imposes a strict, dynamically evolving kinematic ceiling $M_H(z)$, offering a distinct observational discriminator from standard PBH scenarios.

3.3 Characteristic Mass Scale, Probability of Saturation, and Redshift Dependence

The dimensional saturation scale is

$$M_H(z) \equiv \frac{c^3}{GH(z)}, \quad (7)$$

which is the natural mass scale associated with a causally coherent region at redshift z (the Hubble-horizon mass). The null-kinematic claim is not that an entire Hubble patch collapses, but that *saturation is a rare-event threshold* in the local null-directional statistics: only exceptional regions reach the polarization/strain required for kinematic saturation.

Accordingly, instead of introducing a fixed “efficiency” factor, the prediction is parameterized in terms of a *saturation probability* that depends on scale and epoch:

$$P_{\text{sat}}(M, z) \equiv \text{Pr}[\text{local null ensemble reaches saturation on mass scale } M \text{ at } z]. \quad (8)$$

The resulting SMBH abundance is then governed by the tail of this probability, rather than by baryonic accretion histories.

A convenient phenomenological representation is a large-deviation (rate-function) form,

$$P_{\text{sat}}(M, z) \sim \exp[-\mathcal{S}(M, z)], \quad (9)$$

where $\mathcal{S}(M, z)$ summarizes the rarity of the required null-directional polarization/strain excursion seeded by primordial fluctuations. This makes explicit that the smallness previously absorbed into an “efficiency” parameter is instead a statistical statement about the extreme tail of the primordial field. In inflationary cosmology, $\mathcal{S}(M, z)$ would be expected to depend on the variance of primordial curvature perturbations at the corresponding scale, linking the abundance of early SMBHs directly to the statistics of the initial power spectrum.

Minimal tail model from null-solid-angle strain. In the null-directional formulation, gravitational structure corresponds to a local deficit in accessible null solid angle. This deficit is quantified by the scalar strain

$$\mathcal{I}(x) \equiv \ln\left(\frac{4\pi}{\Omega(x)}\right), \quad (10)$$

where $\Omega(x) \leq 4\pi$ denotes the accessible null solid angle of the local directional ensemble and a coarse-grained strain $\mathcal{I}_R$ is defined on smoothing scale $R(M)$. Kinematic saturation is modeled as a threshold event,

$$\text{saturation occurs if } \mathcal{I}_R > \mathcal{I}_*, \quad (11)$$

where $\mathcal{I}_*$ is a theory-native critical polarization/strain threshold.

If the primordial field sourcing $\mathcal{I}_R$ fluctuations is approximately Gaussian on the relevant large smoothing scales (as expected for linear-regime curvature perturbations and weakly non-linear coarse-graining),

$$P_{\text{sat}}(M, z) = \text{Pr}(\mathcal{I}_R > \mathcal{I}_*) \simeq \frac{1}{2} \text{erfc}\left(\frac{\mathcal{I}_*}{\sqrt{2} \sigma_{\mathcal{I}}(M, z)}\right), \quad (12)$$

where $\sigma_{\mathcal{I}}(M, z)$ is the rms variance of $\mathcal{I}_R$ on mass scale M at epoch z . In the weak-strain regime, minimal linear closure is adopted by taking the strain $\mathcal{I}_R$ to be locally proportional to the coarse-grained linear matter density contrast, δ_R , such that $\mathcal{I}_R = A\delta_R$. Operationally, the relevant convergent-flow excursions are treated as being sourced by the same linear perturbation field; in linear theory, the velocity divergence and density contrast are proportional ($\nabla \cdot \mathbf{v} \propto -aHf\delta$), so modeling saturation as a threshold in δ_R provides a minimal proxy for the flow-based trigger. The scalar strain variance is thus $\sigma_{\mathcal{I}}(M, z) = A\sigma_R(M, z)$, where σ_R is the standard Λ CDM mass variance computed from the primordial power spectrum and transfer function, and A is an order-unity mapping constant.

Constitutive interpretation of the strain variable. In the broader NDK constitutive framework [2], the scalar strain is more fundamentally defined as the relative entropy of the local null-directional distribution $P(\hat{\mathbf{k}}; x)$ relative to isotropy,

$$I[P](x) = \int P(\hat{\mathbf{k}}; x) \ln(4\pi P(\hat{\mathbf{k}}; x)) d\Omega.$$

The null-solid-angle quantity $\mathcal{I}(x) \equiv \ln(4\pi/\Omega(x))$ used in the present work should therefore be understood as a reduced phenomenological proxy for this more general strain functional, appropriate when the local directional state is summarized primarily by its net support deficit. In the weak-anisotropy regime, the constitutive expansion takes the schematic form

$$I \approx \frac{3}{2} m^2 + \frac{15}{4} Q^2,$$

where m denotes the dipolar transport bias and Q the quadrupolar anisotropy of the local null ensemble. This indicates that saturation-driving strain need not arise solely from isotropic overdensity, but may also receive contributions from directional transport bias and anisotropic tidal deformation. The present paper nevertheless retains the minimal density-based closure $\mathcal{I}_R = A\delta_R$ as the simplest observational proxy for constructing a falsifiable cosmological envelope.

In the rare-tail regime $\mathcal{I}_* \gg \sigma_{\mathcal{I}}$,

$$P_{\text{sat}}(M, z) \approx \frac{\sigma_{\mathcal{I}}(M, z)}{\sqrt{2\pi} \mathcal{I}_*} \exp\left[-\frac{\mathcal{I}_*^2}{2\sigma_{\mathcal{I}}^2(M, z)}\right]. \quad (13)$$

Thus the small formation fraction is controlled by a single threshold $\mathcal{I}_*$ together with the fluctuation amplitude $\sigma_{\mathcal{I}}(M, z)$ determined by the primordial power spectrum on the corresponding scale. Eq. (12) provides an explicit minimal realization of the rate-function form in Eq. (9).

Implied mass function scaling. The comoving number density of saturation-formed black holes may be estimated in direct analogy with threshold-based structure formation models [7, 8]. If regions of mass scale M are coarse-grained on smoothing scale $R(M)$ with background mass density $\bar{\rho}(z)$, then the differential abundance may be written schematically as

$$\frac{dn}{dM}(M, z) \sim \frac{\bar{\rho}(z)}{M} \left| \frac{dP_{\text{sat}}(M, z)}{dM} \right|. \quad (14)$$

While the mathematical structure parallels Press–Schechter threshold statistics, the physical trigger differs: standard excursion-set theory models nonlinear gravitational collapse of overdensities exceeding a density contrast threshold, whereas the present framework models a kinematic saturation of null-directional strain that does not require hydrostatic instability or pressure-overcoming collapse.

Using the rare-tail form of Eq. (13), this yields the leading exponential scaling

$$\frac{dn}{dM}(M, z) \propto \exp\left[-\frac{\mathcal{I}_*^2}{2\sigma_{\mathcal{I}}^2(M, z)}\right], \quad (15)$$

up to algebraic prefactors arising from the scale dependence of $\sigma_{\mathcal{I}}(M, z)$.

Thus, in the null-kinematic framework, the high-mass cutoff of the early SMBH population is controlled by the scale dependence of the null-directional strain variance rather than by accretion timescales. The redshift evolution of the abundance follows directly from the growth and scale dependence of $\sigma_{\mathcal{I}}(M, z)$ inherited from the primordial power spectrum and transfer function.

In particular, because $\sigma_{\mathcal{I}}(M, z)$ decreases toward larger mass scales, the mass function exhibits an exponential suppression near the kinematic support boundary $M \lesssim M_H(z)$, providing a quantitative discriminator from purely accretion-driven growth models. Future surveys measuring the slope of the high-redshift SMBH mass function therefore directly constrain the threshold parameter $\mathcal{I}_*$ and the scale dependence of $\sigma_{\mathcal{I}}(M, z)$.

Observable inference strategy. The null-kinematic prediction may be confronted with high-redshift SMBH surveys through a minimal inference procedure in which the rare-event contour level p and the effective threshold combination $\mathcal{I}_*/A$ are constrained by the observed high-mass tail. For a fixed background cosmology, the smoothed variance $\sigma_R(M, z)$ on mass scale M is computed from the primordial power spectrum and transfer function. In the weak-strain regime, the variance may be written as $\sigma_{\mathcal{I}}(M, z) = A \sigma_R(M, z)$, where A is an order-unity calibration factor relating null-directional strain to the coarse-grained fluctuation amplitude.

In the large-deviation limit, the resulting mass function exhibits exponential suppression,

$$\frac{dn}{dM} \propto \exp \left[-\frac{\mathcal{I}_*^2}{2A^2\sigma_R^2(M, z)} \right].$$

In principle, fitting the high-mass tail in a given redshift bin constrains the effective combination $\mathcal{I}_*/A$, after which the model predicts the redshift evolution of the cutoff without additional parameters. Consistency of this evolution across redshift therefore provides a direct empirical test of the saturation-threshold hypothesis. The absolute upper limit of collapse remains bounded independently by the causal kinematic scale $M_H(z)$.

For representative early-universe expansion rates ($z \sim 10\text{--}15$), the Hubble parameter is $H(z) \sim 10^{-17}\text{--}10^{-16} \text{ s}^{-1}$. Observed SMBH masses may be interpreted as lying on a rarity-weighted contour within this support, defined implicitly by

$$P_{\text{sat}}(M_p(z), z) = p. \quad (16)$$

In this representation, SMBH-scale objects correspond to extreme-tail formation within the kinematic boundary $M \lesssim M_H(z)$ rather than collapse of an entire Hubble-coherent region.

Calibrated envelope and observational comparison. For visualization, a calibrated rare-event envelope is constructed by first defining a constant-probability contour through

$$P_{\text{sat}}(M_p(z), z) = p,$$

with representative rarity level $p = 10^{-9}$, and then imposing the absolute causal support limit

$$M_{\text{env}}(z) = \min[M_{\text{contour}}(z), M_H(z)].$$

The effective threshold combination $\mathcal{I}_*/A$ is phenomenologically calibrated by requiring the underlying constant- p contour to lie one order of magnitude above the observed mass of J0313–1806 at $z = 7.642$ [9], thereby defining a conservative upper-capacity bound rather than a median fit. The resulting envelope lies above the currently reported population of high-redshift objects, including GN-z11 [3], UHZ1 [4, 5], and J1342+0928 [10], while naturally expanding toward later epochs as the absolute kinematic support scale $M_H(z)$ increases with decreasing $H(z)$ (Figure 1). Because this envelope is calibrated as a conservative upper-capacity bound rather than a median fit, modest future exceedances would indicate tension in the adopted realization, whereas statistically robust exceedances beyond the quoted calibration and mass-estimate uncertainties would constitute falsification. For comparison, standard Eddington-limited accretion tracks from direct-collapse and Pop III seed masses are also shown.

Kinematic support boundary + rare-event contour (Planck18 params; Colossus linear variance; calibrated ceiling)

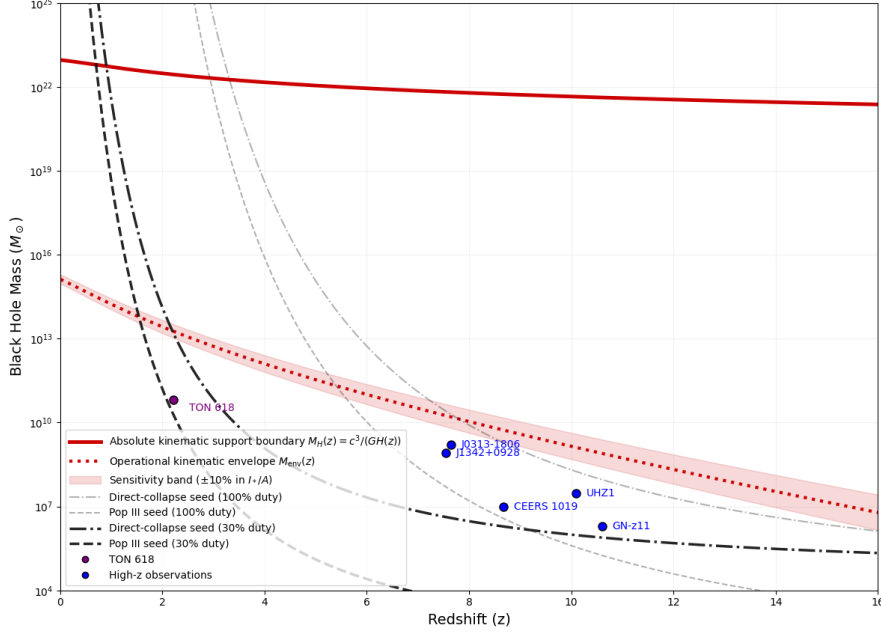


Figure 1: **Kinematic Support Boundary and Calibrated Rare-Event Envelope.** The solid red curve shows the absolute kinematic support boundary $M_H(z) = c^3/(GH(z))$. The dotted red curve shows the operational kinematic envelope $M_{\text{env}}(z)$ obtained from a constant-probability rare-event construction with $P_{\text{sat}} = p$ and $p = 10^{-9}$, clipped by the causal support limit such that $M_{\text{env}}(z) = \min[M_{\text{contour}}(z), M_H(z)]$. The shaded red band shows the sensitivity of this envelope to a $\pm 10\%$ variation in the calibrated threshold ratio $\mathcal{I}_*/A$. The effective threshold parameter $\mathcal{I}_*/A$ is phenomenologically calibrated by requiring the underlying constant- p contour to lie one order of magnitude ($10\times$) above the observed mass of quasar J0313–1806 [9] at $z = 7.642$, thereby defining a conservative upper-capacity bound rather than a median fit. The variance $\sigma_R(M, z)$ is computed using Planck 2018 Λ CDM cosmological parameters [11] with a linear transfer function (Eisenstein & Hu 1998 [12]) via the Colossus framework [13], assuming a comoving top-hat filter over the mean matter density ρ_{m0} . High-redshift comparison points are shown for reported early accreting black-hole candidates, while a later-epoch comparison point, TON 618 at $z \approx 2.2$ [14], is included to illustrate that the kinematic support scale and calibrated envelope shift upward toward lower expansion rates at later cosmic times. Standard Eddington-limited accretion tracks are plotted for Direct Collapse ($10^5 M_\odot$ at $z = 25$) and Pop III Stellar ($100 M_\odot$ at $z = 30$) seeds under both 30% (solid black) and 100% (faint black) duty cycles.

4 Methodology: Defining the Kinematic Saturation Envelope

This section specializes the phenomenological framework developed above to a concrete observational construction of the upper-capacity envelope, specifying the adopted rarity contour, variance computation, and single-point calibration used in Figure 1.

To test the null-kinematic saturation hypothesis against current high-redshift supermassive black hole (SMBH) observations, a theoretical upper-capacity envelope is constructed and defined as a constant-probability rare-event contour. In the null-directional framework, horizon formation corresponds to a threshold-crossing event in the coarse-grained strain field. Assuming the smoothed strain variable is approximately Gaussian on large cosmological scales, the probability of saturation is modeled by the complementary error-function tail. In the weak-strain mapping, $\sigma_{\mathcal{I}}(M, z) = A\sigma_R(M, z)$, so the saturation threshold may be written in terms of the standard linear variance σ_R as:

$$P_{\text{sat}}(M, z) = \frac{1}{2} \text{erfc}\left(\frac{\mathcal{I}_*}{A\sqrt{2}\sigma_R(M, z)}\right), \quad (17)$$

where $\sigma_R(M, z)$ is the linear-theory rms fluctuation amplitude on smoothing scale $R(M)$ at redshift z , and the combination $\mathcal{I}_*/A$ encodes the effective saturation threshold in units of the smoothed fluctuation variance.

Choice of Rare-Event Contour. Rather than predicting the full mass function normalization,

a constant-probability contour $P_{\text{sat}} = p$ is defined to represent an extreme-tail formation channel. In this work, $p = 10^{-9}$ is adopted as a representative rarity level corresponding to highly exceptional large-scale fluctuation peaks. This choice fixes a threshold variance condition:

$$\sigma_R(M, z) = \frac{\mathcal{I}_*}{A\sqrt{2}\operatorname{erfc}^{-1}(2p)}. \quad (18)$$

For a fixed p , the redshift evolution of the contour is therefore fully determined by the cosmological growth of the linear variance $\sigma_R(M, z)$.

Computation of the Linear Variance. The variance $\sigma_R(M, z)$ is computed within Λ CDM using Planck 2018 cosmological parameters [11]. Linear-theory fluctuations are evaluated using the Colossus framework [13], which implements a standard analytic transfer function (Eisenstein & Hu 1998 [12]) for the linear matter power spectrum. The mass scale M is mapped to a comoving Lagrangian smoothing radius via a spherical top-hat window function over the mean background matter density ρ_{m0} ,

$$R(M) = \left(\frac{3M}{4\pi\rho_{m0}} \right)^{1/3}. \quad (19)$$

The redshift dependence of the variance follows from the linear growth factor $D(z)$.

Calibration of the Effective Threshold. The combination $\mathcal{I}_*/A$ is a theory-native parameter that encapsulates both the microphysical saturation threshold and the mapping between null-directional strain and the coarse-grained density fluctuation amplitude. In the absence of a first-principles derivation of this mapping, the envelope is phenomenologically calibrated using the well-measured high-redshift quasar J0313–1806 ($1.6 \times 10^9 M_\odot$ at $z = 7.642$) [9].

To define an upper-capacity envelope rather than a median fit, the rare-event contour is required to lie exactly one order of magnitude ($10\times$) above this observed mass at the same redshift. This choice defines a conservative capacity bound while avoiding calibration to the precise observed mass value. Solving Eq. (18) at the anchor redshift for the adopted $p = 10^{-9}$ contour and the $10\times$ J0313–1806 calibration prescription yields an effective threshold ratio

$$\frac{\mathcal{I}_*}{A} \approx 3.28.$$

Testing alternative safety margins (e.g., anchoring at $5\times$ or $20\times$) shifts the calibrated threshold $\mathcal{I}_*/A$ by less than $\sim 15\%$, demonstrating that the exponential redshift scaling of the envelope is robust to the exact choice of anchor buffer.

An uncertainty band of $\pm 10\%$ in $\mathcal{I}_*/A$ is included to reflect systematic uncertainties in high-redshift SMBH mass estimates and in the phenomenological strain–variance mapping.

Absolute Kinematic Support Boundary. Independently of the rare-event contour, the null-kinematic framework predicts an absolute causal support scale given by

$$M_H(z) = \frac{c^3}{GH(z)}, \quad (20)$$

which corresponds to the mass contained within a Hubble-coherent region. The final kinematic capacity envelope is therefore defined as the minimum of the rare-event contour and the causal support limit,

$$M_{\text{env}}(z) = \min[M_{\text{contour}}(z), M_H(z)]. \quad (21)$$

Interpretation. With this single-point calibration, the model produces a dynamically evolving envelope whose redshift dependence follows directly from linear growth at fixed rarity level p . The resulting boundary provides an upper-capacity condition for direct kinematic saturation

events and may be contrasted with standard Eddington-limited accretion tracks under typical duty-cycle assumptions. Note that this envelope represents an upper-capacity bound for direct, instantaneous formation; subsequent secular accretion can occur for objects below this bound, but is not explicitly modeled here.

5 Discussion and Conclusion

The null-kinematic framework developed here replaces geometric collapse with a statistical threshold phenomenon: black hole formation corresponds to a rare-event saturation of the local null-directional ensemble. In this picture, supermassive black holes (SMBHs) at high redshift are not the product of prolonged accretion histories alone, but may arise from extreme-tail polarization events governed by primordial fluctuation statistics.

Two distinct and falsifiable predictions emerge:

1. A **hierarchy of kinematic mass scales**: an absolute causal support boundary set by the horizon-scale mass

$$M_H(z) = \frac{c^3}{GH(z)},$$

together with a variance-controlled rare-event capacity envelope

$$M_{\text{env}}(z) \leq M_H(z),$$

which defines the practical upper bound for direct saturation-driven formation under the adopted statistical closure.

2. An **exponential high-mass suppression** of the SMBH mass function,

$$\frac{dn}{dM} \propto \exp\left[-\frac{\mathcal{I}_*^2}{2\sigma_{\mathcal{I}}^2(M, z)}\right],$$

arising from the Gaussian-tail structure of the underlying null-directional strain field. While standard halo mass functions also exhibit exponential suppression, in the present framework the cutoff is controlled by a null-directional strain threshold rather than by non-linear gravitational collapse dynamics, and its redshift evolution is fixed once the threshold parameter is calibrated.

The first prediction therefore defines two logically distinct falsifiers. The absolute boundary $M_H(z)$ constitutes a hard kinematic ceiling independent of statistical assumptions. In practice, however, realistic fluctuation statistics rarely approach this causal limit, and the operative observational discriminator is the calibrated rare-event envelope $M_{\text{env}}(z)$.

With the rarity level p and threshold ratio $\mathcal{I}_*/A$ fixed at $z = 7.642$, the model contains no additional freedom in its redshift evolution. Because the envelope is itself defined as a conservative upper-capacity construction, isolated mild exceedances would place the model under tension. By contrast, a statistically robust population of high-redshift black hole mass estimates lying above the predicted envelope after propagation of the quoted calibration band and observational mass-estimate uncertainties would constitute falsification of the adopted single-anchor Gaussian-tail saturation-threshold realization.

The second prediction links the abundance slope of early SMBHs to the scale dependence of the primordial fluctuation spectrum through $\sigma_{\mathcal{I}}(M, z)$, providing an observational bridge between high-redshift black hole statistics and inflationary initial conditions.

Table 1: Comparison of Phenomenological Implications: Standard Cosmology vs. Null-Kinematic Framework

Observable	Standard GR / Cosmology	Null-Kinematic Prediction
High-z SMBH Formation	Growth from stellar seeds limited by Eddington accretion rates; likely requires very massive seeds and/or prolonged super-Eddington phases to explain the most distant reported accreting black-hole candidates.	Rare-event null-directional saturation when coarse-grained strain $\mathcal{I}_R$ exceeds a critical threshold $\mathcal{I}_*$. Abundance governed by Gaussian-tail statistics of primordial fluctuations.
High-Mass Cutoff	Upper masses determined by integrated growth time and feedback; no intrinsic horizon-scaled boundary.	Kinematic support boundary $M \lesssim M_H(z) = c^3/(GH(z))$ with exponential suppression near the boundary.
Singularity Structure	Geometric singularity ($R \rightarrow \infty$) indicating breakdown of classical description.	Finite saturation limit: curvature is associated with maximal null-directional strain, suggesting an alternative description of singular behavior rather than divergent geometric invariants.

Unlike standard accretion-driven growth models, where the upper mass distribution is primarily set by integrated growth time and radiative feedback, the null-kinematic framework predicts that the high-mass cutoff reflects a statistical threshold in directional strain. The small formation fraction is therefore not an efficiency parameter but a manifestation of large-deviation statistics in the primordial field.

It is important to emphasize the scope of the present analysis. The work does not attempt a full cosmological re-derivation of structure formation, nor does it modify tested gravitational dynamics in weak- and strong-field regimes. Instead, it introduces a minimal phenomenological extension in which horizon formation admits a threshold-based channel controlled by null-directional polarization. The Gaussian-tail realization adopted here is a conservative assumption consistent with weakly non-linear smoothing; departures from Gaussianity would alter the detailed abundance scaling but preserve the existence of the kinematic support boundary. Within the broader NDK constitutive program, the homogeneous isotropic background is expected to correspond to a vacuum-like limit of the null ensemble, while the present paper isolates the complementary problem of rare localized strain excursions away from that background.

Possible role of rotationally enhanced strain. A further constitutive possibility suggested by recent local analyses of the null-directional substrate [1, 2] is that rotational transport bias can couple positively to quadrupolar strain in aligned configurations. If such coupling persists under cosmological coarse-graining, regions with substantial primordial shear or vorticity could in principle approach the saturation threshold more efficiently than predicted by a purely density-based proxy. Establishing this requires a dedicated cosmological constitutive derivation and is therefore left as a future extension rather than assumed in the minimal realization developed here.

Future observational surveys measuring the high-redshift SMBH mass function and its evolution with redshift can therefore directly constrain the threshold parameter $\mathcal{I}_*$ and the variance scaling $\sigma_{\mathcal{I}}(M, z)$. In this sense, early SMBHs provide a potential empirical window into the statistical microstructure underlying spacetime.

While the present work focuses on cosmological saturation phenomena, the same null-directional strain variable may have implications for microscopic quantum transport and decoherence through finite directional resolution of the substrate. A first-principles derivation of such effects remains an open direction for future research.

Experimental validation of a horizon-scaled support boundary together with a variance-controlled exponential tail would provide empirical evidence that black hole formation may be governed by an underlying statistical constraint on null propagation, rather than purely by classical geometric collapse.

Future directions. A broader cosmological question suggested by the present framework is whether the null-kinematic support scale may also regulate the backward extrapolation of the early universe. Since the causal support boundary scales as $M_H(z) \sim c^3/(GH(z))$, the limit $H \rightarrow \infty$ corresponds not to an arbitrarily large support capacity, but to the collapse of the admissible coarse-grained support mass toward zero. This raises the possibility that, within a fully developed null-directional theory, the standard Big Bang singularity of classical GR may be replaced by a zero-support or maximal-strain boundary at which the effective transport description ceases to be valid. The present work does not derive such a cosmological origin law, nor does it claim a bounce or channel-generation mechanism. However, the existence of a kinematic support ceiling suggests that a non-singular pre-geometric transition may be a natural direction for future development of the framework.

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